Nazmul Huda Badhon Dhaka, Bangladesh
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Profile Summary

Computer Science graduate with two years of research experience in deep learning, machine learning, and computer vision, supported by eight peer-reviewed publications in international journals. Experienced in neural network-based modeling, medical image analysis, and applied AI research. Seeking PhD opportunities in deep learning and AI-driven solutions for real-world problems.

Education

B.Sc. in Computer Science and Engineering — **January 2022 - December 2025**
Daffodil International University, Dhaka, Bangladesh

- CGPA: 3.88/4.00, 85% average
- Thesis: Tabular data-based lightweight convolutional neural network for electricity energy demand prediction.

Exchange Semester in Computer Engineering — **September - December 2023**
Dongseo University, Busan, South Korea

- GPA: 3.94/4.50, 93.25% average

Research Interests

Designing computationally efficient deep neural network architectures.
Developing robust and generalizable deep learning models for real-world applications.

Research Experience

Research Associate — **January 2024 - Present**
Under the supervision of Associate Professor Nazmun Nessa Moon
Department of Computer Science and Engineering,
Daffodil International University, Dhaka, Bangladesh

- Contributed to dataset development, experimentation, and manuscript preparation.
- Supported collaborative academic research and publication activities.
- Research focus: deep learning, medical imaging, and computer vision.

Research Contributor — **December 2025 - March 2026**
8th ABC Challenge, Kyushu Institute of Technology, Japan

- Conducted research on real-world multimodal sensor data from a nursing care facility.
- Developed a Temporal Convolutional Network-based method for indoor location classification using BLE beacon signals.

Research Publications

Journal Articles

1. **Nazmul Huda Badhon**, Imrus Salehin, Md. Tomal Ahmed Sajib, Md. Sakibul Hassan Rifat, S M Noman, Nazmun Nessa Moon. TLCNN: tabular data-based lightweight convolutional neural network for electricity energy demand prediction. *Global Energy Interconnection* (2025), Elsevier. Q1

2. **Nazmul Huda Badhon**, Sabit Ahmed Preanto, Abu Shahed Shah Md Nazmul Arefin, Kazi Jahid Hasan, and Md. Baharul Islam. Temporal convolutional network based indoor location recognition from ble beacon signals in nursing care facility. *International Journal of Activity and Behavior Computing* (2026).

3. Md. Tomal Ahmed Sajib, **Nazmul Huda Badhon**, Imrus Salehin, Md. Sakibul Hassan Rifat, Faysal Ahmmed, Nazmun Nessa Moon. A comparative deep learning methodology for plant insect image classification: assessment of CNN architectures and augmentation techniques. 103680, *MethodsX* (2025). Q2

4. Imrus Salehin, Md. Tomal Ahmed Sajib, **Nazmul Huda Badhon**, Md. Sakibul Hassan Rifat, Nazrul Amin, Nazmun Nessa Moon. Systematic Literature Review of LLM-Large Language Model in Medical: Digital Health, Technology and Applications. 7.9: e70365, *Engineering Reports* (2025). Q2

5. Imrus Salehin, **Nazmul Huda Badhon**, Md Tomal Ahmed Sajib, Mohd Saifuzzaman, and Nazmun Nessa Moon. IQ-UltraRecon: Demodulated IQ ultrasound dataset of human hand and arm tissue for deep learning-based reconstruction. *Data in Brief* (2025), 112001, Elsevier. Q3

6. Imrus Salehin, **Nazmul Huda Badhon**, Md Tomal Ahmed Sajib, and Nazmun Nessa Moon. DeepUS-ReconSeg: A multi-angle paired B-mode Ultrasound dataset for medical imaging reconstruction and segmentation. *Data in Brief* (2025), 112083, Elsevier. Q3

7. Imrus Salehin, Mahbubur Rahman Khan, Ummya Habiba, **Nazmul Huda Badhon**, Nazmun Nessa Moon. BAU-Insectv2: An Agricultural Plant Insect Dataset for Deep Learning and Biomedical Image Analysis. *Data in Brief* (2024), 110083, Elsevier. Q3

Book Chapter:

8. Imrus Salehin, **Nazmul Huda Badhon**, Md. Tomal Ahmed Sajib, Md. Sakibul Hassan Rifat, Nazmun Nessa Moon. Predicting Peak Energy Demand Using Machine Learning Techniques for Efficient Electricity Supply Management. *Taylor & Francis*. Accepted on 24 Apr 2025 (In press).

Projects

Selected Research Projects

1. Breast Cancer Classification using Hybrid Segmentation Deep Learning

- Developed a hybrid U-Net and CNN model.
- Evaluated model performance using accuracy and ROC-based metrics.

2. Lung and Colon Cancer Histopathology Classification

- Applied preprocessing and transfer learning techniques to histopathology images.
- Evaluated classification performance using accuracy and F1-score.

3. Vision Transformer–Based Medical Image Classification

- Implemented a ViT-based classification pipeline for medical image analysis.
- Achieved 87% test accuracy.

Additional Projects

4. Lost and Found Web Application

- Built a full-stack web application for reporting and retrieving lost items.
- Used HTML, CSS, JavaScript, PHP, and MySQL with image upload functionality.

Technical Skills

Programming:	Python, Java, C
AI & ML:	Machine Learning, Deep Learning (DL), Computer Vision, Medical AI
DL Techniques:	CNN, U-Net, Vision Transformers, Transfer Learning
Frameworks:	PyTorch, TensorFlow, Keras, Scikit-learn, OpenCV
Data Analysis:	NumPy, Pandas, Matplotlib, Seaborn
Research Tools:	GitHub, Google Colab, LaTeX

Language Proficiency

- Bangla: Native
- English: Professional working proficiency

Academic Service

- Peer Reviewer, Digital Health, SAGE Publishing, 2025

Awards and Achievements

- Research Award for Scholarly Publication in Scopus-Indexed Journals, Division of Research, Daffodil International University, 2024, 2025
- Best Performer Award, Computer Vision and Medical Image Analysis Bootcamp, Health Informatics Research Lab, Daffodil International University, 2025
- Global Korea Scholarship (fully funded), Government of the Republic of Korea, 2023
- Merit-based Scholarship, Daffodil International University, 2022-2025
- Government General Board Scholarship, Education Board, Jashore, 2015

Leadership and Extracurricular Activities

Team Leader **2023**
International Camp, Dongseo University, South Korea

Volunteer Intern **2024**
International Affairs, Daffodil International University

Founding General Secretary **March 2024 - July 2024**
DIU Research Society

References

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